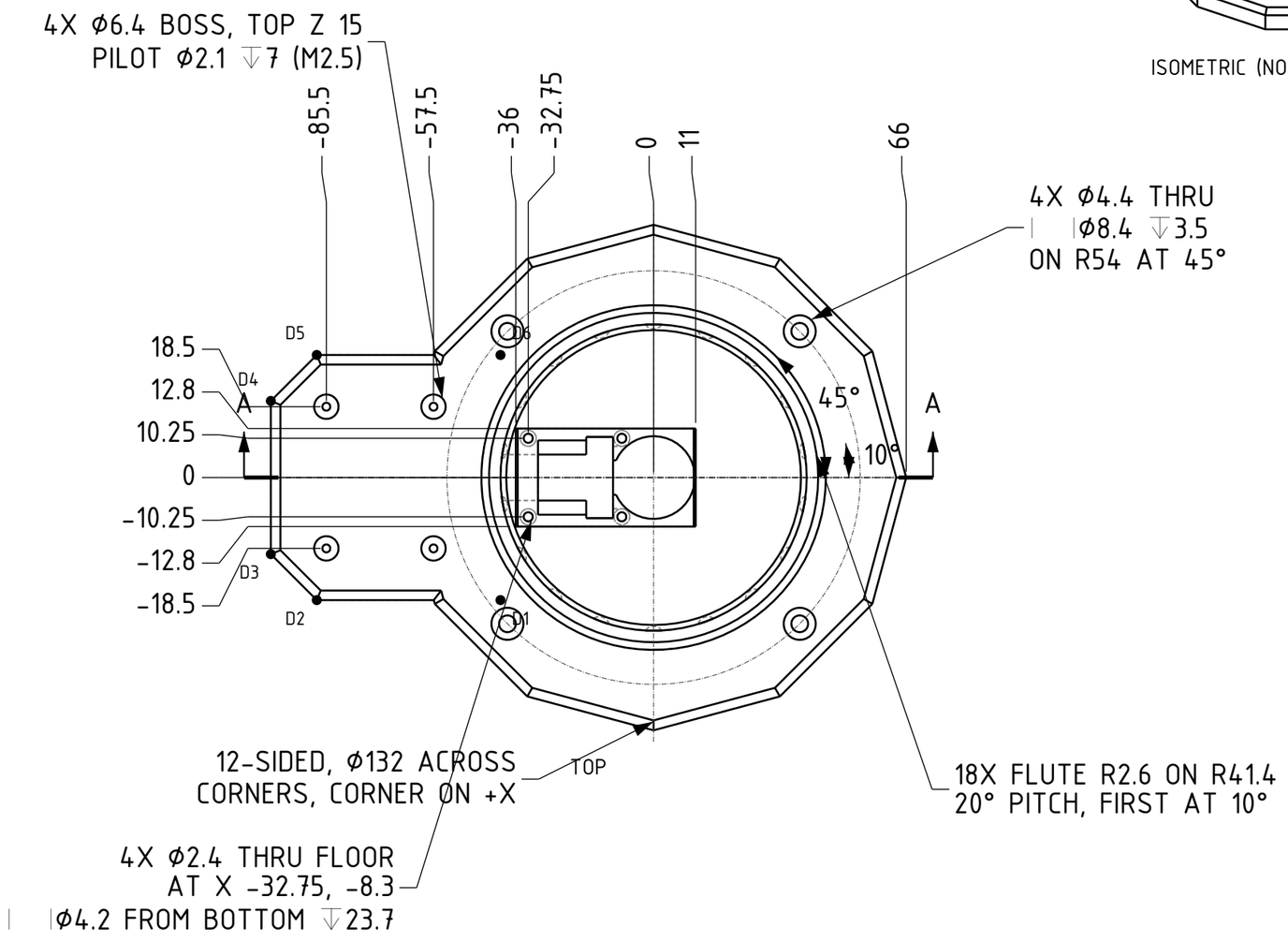
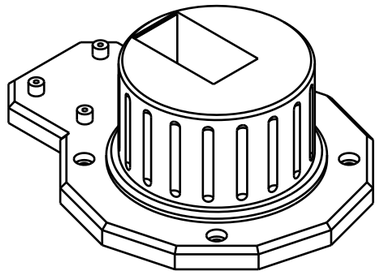


ISOMETRIC (NOT TO SCALE)

## BOARD DECK POINTS (X, Y)

| PT | X | Y |
|---|---|---|
| D1 | -40 | -32 |
| D2 | -88 | -32 |
| D3 | -100 | -20 |
| D4 | -100 | 20 |
| D5 | -88 | 32 |
| D6 | -40 | 32 |

## FEATURES (in order)

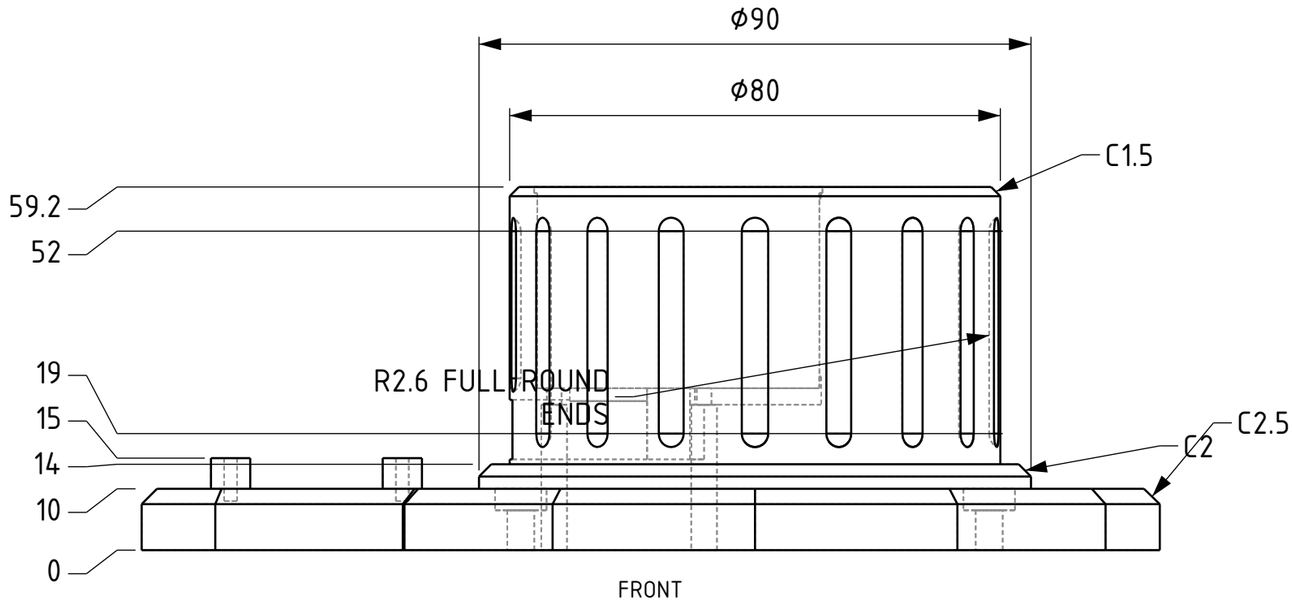
1. Plinth: regular 12-gon, ⌀132 across corners (a corner on +X), z 0 to 10.
2. Board deck: polygon D1 to D6, z 0 to 10; union with the plinth.
3. Chamfer C2.5 on all top outer edges (z 10).
4. Mounting holes: 4X ⌀4.4 through on R54 at 45° + 90°n; counterbore ⌀8.4 from z 6.5 up.
5. Collar ⌀90, z 10 to 14, C2 on its top edge.
6. Column ⌀80, z 10 to 59.2, C1.5 on its top edge.
7. Board standoffs: 4X ⌀6.4, z 9 to 15, at x -57.5 / -85.5, y ±18.5; pilot ⌀2.1 from z 8 to 16.
8. Flutes: 18X vertical ⌀5.2 on R41.4 at 10° + 20°n, axis z 19 to 52 (z 28 to 52 for the two at 170° and 190°, over the cable tunnel), spherical ends; cut.
9. Pocket P (sheet TAL-001) with its servo frame origin P0 at (0, 0, 43.3): servo +Z (shaft) along +Z, servo +X along +X, servo +Y along +Y.
10. Top opening: cut box x -36 to 11, y ±12.8, from z 58.2 up (the way in: the servo drops in from the top, connector end first).
11. Cable tunnel (plug bay to board deck): cut box x -60 to -13, y ±6, z 14.8 to 24.5. Pocket P's bump channel (z down to 24.3) opens into its roof.
12. Case screw holes from below (TAL-002, seat 19.6): 4X at x -32.75 and x -8.3, y ±10.25: ⌀2.4 from z 23.7 up to the pocket floor (z 26.4), ⌀4.2 counterbore from the bottom to z 23.7. The pair at x -8.3 runs 0.2 from the plug bay: cut the wall between them away (box x -10.6 to -8.3, |y| 8.15 to 10.6, z 14.8 to 23.7).

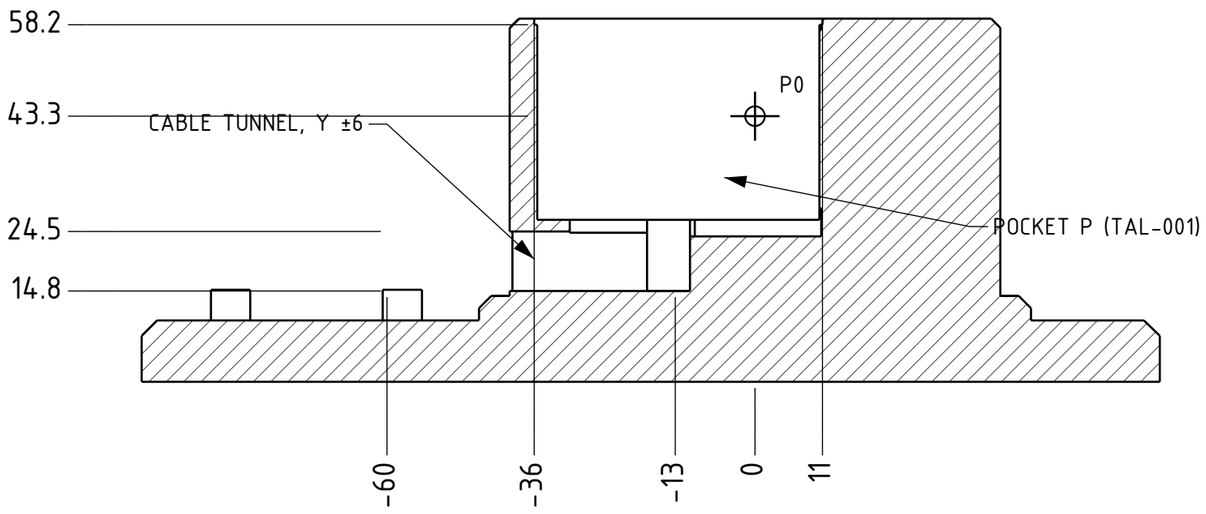
## NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- The pan servo drops in from the top (its horns may stay on); its plugs hang into the bay under it. 4X M2 × 6 case screws from below, two at each end of the servo, so the arm's overturning moment can't rock it in the pocket.
- Print: plinth down; no support.

THIRD ANGLE PROJECTION

| WORK SET A: base, shoulder, upper arm, hub caps, fit gauge | |
|---|---|
| PROJECT<br>TALOS 5-axis arm | EST. MASS<br>154 g |
| PART<br>Base (part sheet 1 of 2) | SCALE<br>2:3 |
| PART NO.<br>TAL-101 | UNITS / REV<br>mm / A |
| QTY / ARM<br>1 | DATE<br>2026-11-10 |
| MATERIAL<br>PLA, 20 % infill, 3 perimeters | SHEET<br>6 of 21 |




SECTION A-A

SECTION A-A

Section on the XZ plane (cutting line on sheet 1), looking toward +Y.

P0 = pocket P servo frame origin; servo axes along the part axes.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

| WORK SET A: base, shoulder, upper arm, hub caps, fit gauge | |
|---|---|
| PROJECT<br>TALOS 5-axis arm | EST. MASS<br>- |
| PART<br>Base (part sheet 2 of 2) | SCALE<br>1:1 |
| PART NO.<br>TAL-101 | UNITS / REV<br>mm / A |
| QTY / ARM<br>1 | DATE<br>2026-11-10 |
| MATERIAL<br>PLA, 20 % infill, 3 perimeters | SHEET<br>7 of 21 |